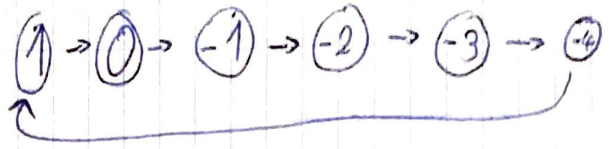
Zadanie 1.



Q_A	Q_B	Q_C	Q_A^t	Q_B^t	Q_C^t	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	1	1	1	1	1	1	1	1	1
0	0	1	0	0	0	0	0	0	0	1	1
0	1	0	0	0	1	0	0	1	1	1	1
0	1	1	0	0	0	0	0	0	0	1	1
1	0	0	0	0	1	1	1	0	0	1	1
1	0	1	1	0	0	0	0	0	0	1	1
1	1	0	1	0	1	0	0	1	1	1	1
1	1	1	1	1	0	0	0	0	0	1	1

$J = K$

J_A

Q_A \ $Q_B Q_C$	00	01	11	10
0	1	0	X	X
1	1	0	0	0

$\overline{Q_B} \cdot \overline{Q_C}$

J_B

Q_A \ $Q_B Q_C$	00	01	11	10
0	1	0	X	X
1	0	0	0	1

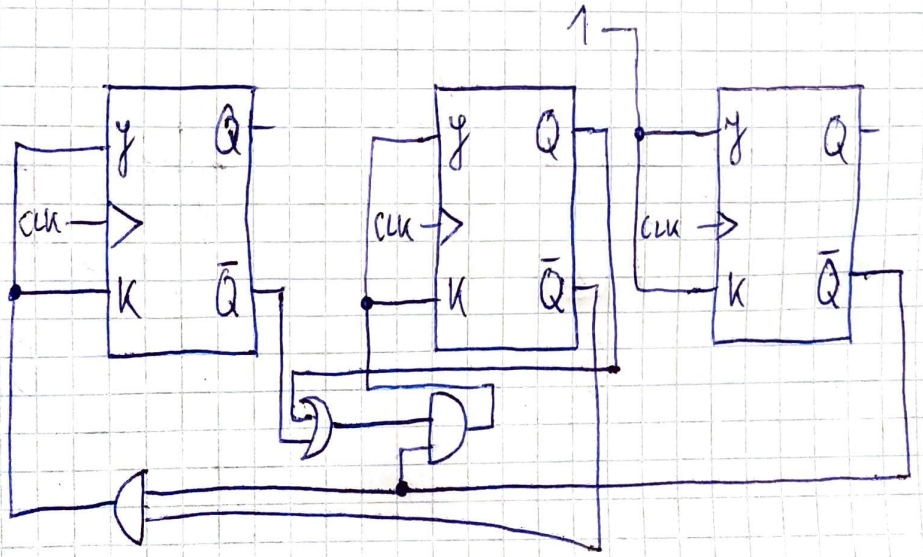
$\overline{Q_C} \cdot (\overline{Q_A} + Q_B)$

J_C

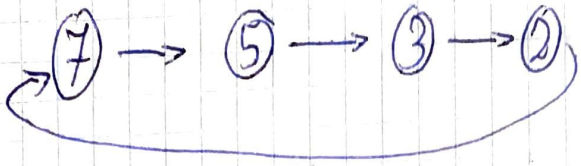
Q_A \ $Q_B Q_C$	00	01	11	10
0	1	1	X	X
1	1	1	1	1

1

Układ.



zadanie 2.



Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	S_A	R_A	S_B	R_B	S_C	R_C
0	0	0	0	1	1	0	1	1	0	1	0
0	0	1	0	1	0	0	1	1	0	0	1
0	1	0	0	1	1	1	0	0	0	1	0
0	1	1	1	0	0	0	0	0	0	0	1
1	0	0	0	1	1	0	1	1	0	1	0
1	0	1	0	1	0	0	1	1	0	0	1
1	1	0	1	0	1	1	0	0	1	1	0
1	1	1	1	0	1	0	0	0	1	0	0

S_A

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	0	1
1	X	0	0	X

$\overline{Q_C}$

S_B

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	0	0
1	X	1	0	X

$\overline{Q_B}$

S_C

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	0	1
1	X	0	0	X

$\overline{Q_C}$

R_A

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	0	0
1	X	1	0	X

$\overline{Q_B}$

R_B

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	0	0
1	X	0	1	X

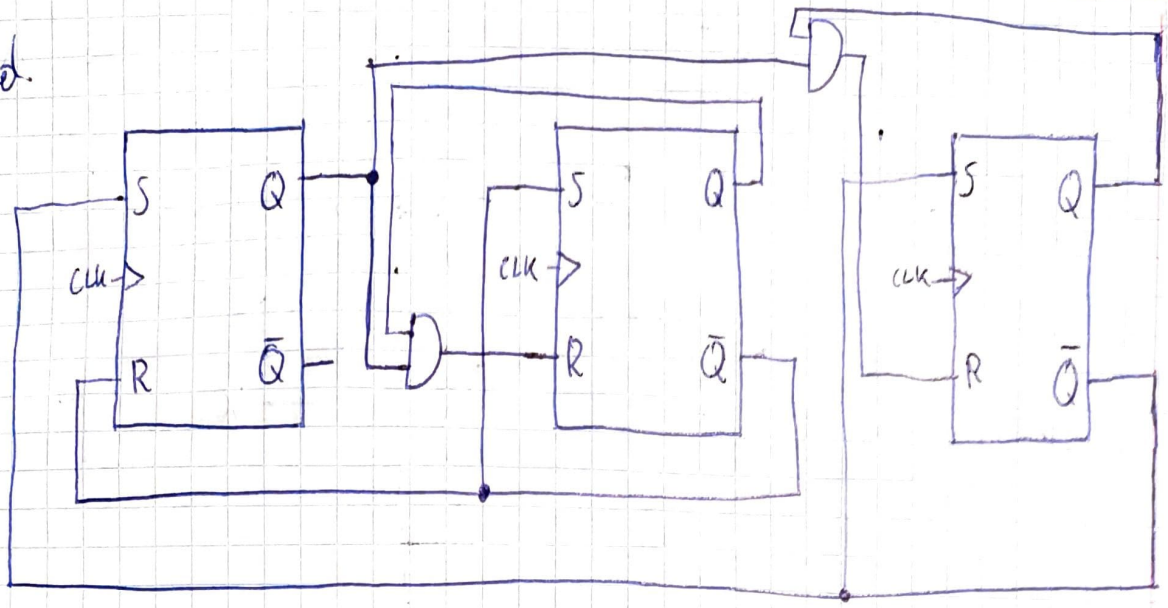
$Q_A \cdot Q_B$

R_C

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	1	0
1	X	0	0	X

$Q_A \cdot Q_C$

układ.



Zadanie 3

$0 \to 1 \to 2 \to 1 \to 2 \to 3 \to 0$

000 001 010 101 110 011 000

$J = K$

Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	0	1	0	0	0	0	1	1
0	0	1	0	1	0	0	0	1	1	1	1
0	1	0	1	0	1	1	1	1	1	1	1
0	1	1	0	0	0	0	0	1	1	1	1
1	0	0	1	0	1	0	0	0	0	1	1
1	0	1	1	1	0	0	0	1	1	1	1
1	1	0	0	1	1	1	1	0	0	1	1
1	1	1	1	0	0	0	0	1	1	1	1

J_A

Q_A \ $Q_B Q_C$	00	01	11	10
0	0	0	0	1
1	x	0	x	1

$Q_B \cdot \bar{Q}_C$

J_B

Q_A \ $Q_B Q_C$	00	01	11	10
0	0	1	1	1
1	x	1	x	0

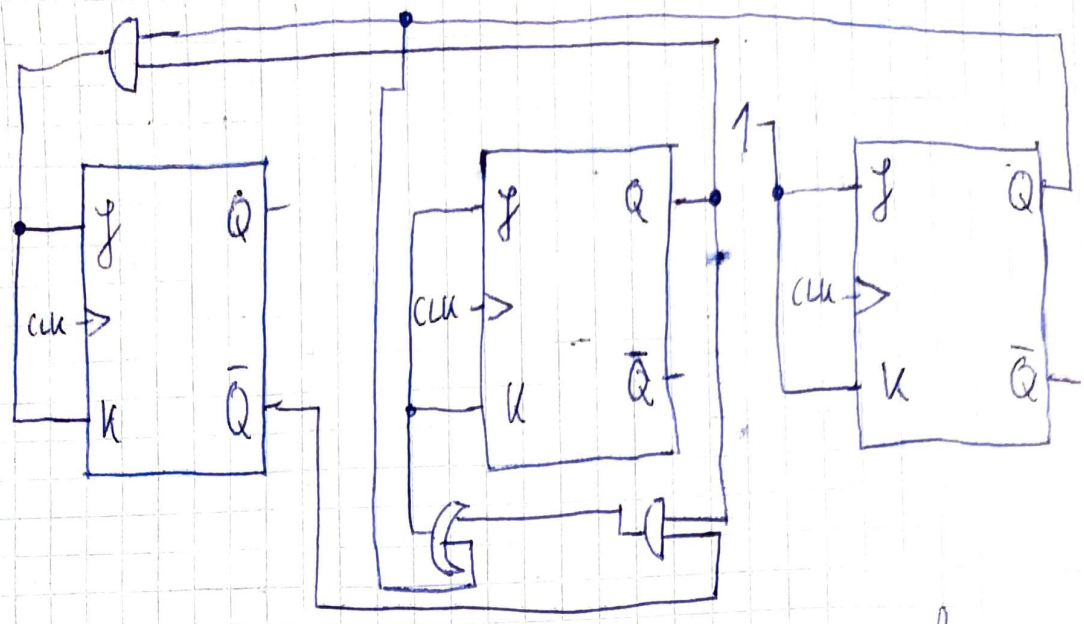
$Q_C + Q_B \cdot \bar{Q}_A$

J_C

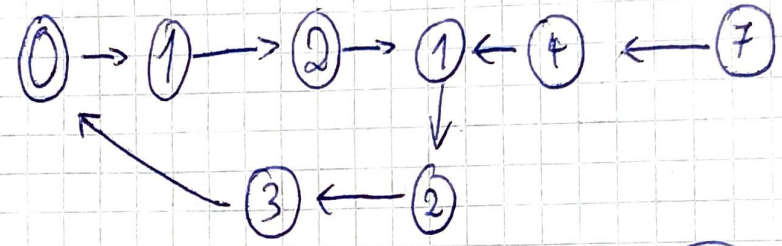
Q_A \ $Q_B Q_C$	00	01	11	10
0	1	1	1	1
1	x	1	x	1

1

układ.



trajektoria.



Licznik jest bezpieczny, bo zawsze wraca do pierwotnej trajektorii.

zadanie 4.

X	Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	T_A	T_B	T_C
0	0	0	0	0	1	0	0	1	0
0	0	0	1	1	1	0	1	1	1
0	0	1	0	1	0	0	1	1	0
0	0	1	1	1	0	1	1	1	1
0	1	0	0	1	1	0	1	1	0
0	1	0	1	0	0	0	1	1	1
0	1	1	0	0	0	0	1	1	0
0	1	1	1	0	0	1	1	1	1
1	0	0	0	0	1	1	0	1	1
1	0	0	1	0	1	0	0	1	0
1	0	1	0	1	0	1	1	1	1
1	0	1	1	1	0	0	1	1	0
1	1	0	0	1	1	1	1	1	1
1	1	0	1	1	1	0	1	1	0
1	1	1	0	1	1	1	1	1	1
1	1	1	1	0	0	1	0	0	0

$X Q_A$ \ $Q_B Q_C$	00	01	11	10
00	0	1	1	1
01	0	1	1	1
11	1	0	1	1
10	1	0	1	1

$$T_A = X \oplus Q_C + Q_B$$

$$T_B = 1$$

$X Q_A$ \ $Q_B Q_C$	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$$T_C = X \oplus Q_C$$

